

José Pablo Rodríguez Pinal

Mechatronics Engineering Student

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Education

Instituto Tecnológico y de Estudios Superiores de Monterrey - Campus Monterrey

Bachelor of Science in Mechatronics Engineering (B.S.)

Expected Graduation: Dec 2027

Professional Profile

Mechatronics Engineering student with hands-on experience in industrial automation and electromechanical control systems. I have implemented solutions using Siemens SIMATIC S7 PLCs on real hardware, integrating HMI systems, embedded systems (STM32), and computer vision. Seeking to join as an intern in automation and control areas with an industrial focus.

Major Projects

Industrial Automation with Siemens SIMATIC PLC and HMI Development

October 2025

- PLC programming using SIMATIC S7 in TIA Portal for a physical industrial sorting system with magnetic sensors.
- Development of an HMI for remote crane-type control with real-time monitoring and positioning on a functional prototype.

Partner Institution: National Institute of Astrophysics, Optics and Electronics (INAOE)

Design, Modeling, and Control of Assistive Robotic Manipulator with Computer Vision

January 2026

- Dynamic modeling (double pendulum approach) and implementation of PD control with gravity compensation for safe motion.
- Integration of computer vision (YOLO, Python) for object detection and automatic trajectory execution in RoboDK.

Partner Institution: John Deere

Design and Implementation of a Mechatronic Automation System for a Scaled Tractor

December 2026

- CAD design in SolidWorks and electromechanical integration of a scaled tractor using STM32, IMU, encoders and ultrasonic sensor.
- Implemented trajectory and speed control using fuzzy logic.
- Developed a MATLAB interface for parametrized trajectory execution with real-time telemetry transmission (speed, distance, orientation and obstacle detection).

Partner Institution: Kemet

Industrial Vision System for Quality Inspection

May 2026

- Development of an automated system using ESP32-CAM for detection of printing defects in a manufacturing process.
- Automation of process stop upon error detection, reducing material waste and improving inspection time.

Technical Skills

PLC: Siemens SIMATIC S7 (TIA Portal), Allen-Bradley (Studio 5000, Ladder Programming)

Software: Emulate3D, MATLAB/Simulink, FluidSIM, SolidWorks, Python, RoboDK.

Embedded Systems: STM32, ESP32, Arduino

International Distinctions

Bronze Medal – ISSSP International Nano-Satellite Competition 2024, Hong Kong

Languages

Spanish: Native

English: Advanced – TOEFL ITP 590

German: B2 Goethe-Institut Certified